

64× NVIDIA B200 AI Inference Cluster Specification

Rev 1.3 · 2026-07-19

Generated from Markdown — edit the `.md` source and regenerate (`python3 docs/md2pdf.py`)

Contents

1	Design Goals	4
2	Physical Summary	5
3	Rack Layout and Facility Zones	6
3.1	How many racks?	6
3.2	Why a dedicated BOOT rack?	6
3.3	Why one STOR rack (usually)?	7
3.4	Baseline elevation (logical)	7
3.5	Per-rack contents (RU budget sketch)	7
3.6	Decision summary	8
4	Compute Nodes	9
4.1	GPU Server Platform	9
4.2	Per-Server Specification	9
4.3	Cluster Aggregate Capacity	9
5	GPU Parallelism Model	11
5.1	Logical Layout	11
5.2	Parallelism Dimensions	11
5.3	Example Configurations	11
6	Network Architecture	12
6.1	Host NIC Model — 1× 400GbE per GPU (required baseline)	12
6.2	Fabric Requirements	12
6.3	Topology Rules	13
6.4	Topology — Rail-Optimized (default)	13
6.5	Leaf Switches — Arista 7060DX5-32 (rail leaves)	14
6.6	Spine Switches — 7060DX5-64 (qty 2)	15
6.7	Server Connectivity (per GPU server)	15
6.8	Port Plan (Planner Baseline)	15
6.9	Optics and Cabling	16
6.10	RoCEv2 / AI Ethernet Profile (Arista EOS)	16
6.11	Out-of-Band / Management Network	16
7	Storage	19
7.1	Role of Storage in This Cluster	19
7.2	Logical Architecture	19
7.3	Integration with the Network	20
7.4	Capacity and Performance Targets	20
7.5	Node-Local Storage (already on GPU servers)	22
7.6	Kubernetes Integration	22
7.7	Inference Capture and 1-Year Archive Pipeline	23
7.8	Data Layout	24
7.9	Platform Options	24
7.10	Reference Hardware Footprint	25
7.11	Failure and Lifecycle	25
7.12	What Storage Does <i>Not</i> Do Here	25
7.13	Summary Integration Map	25
8	Kubernetes Architecture	27
8.1	Management Nodes	27
8.2	GPU Workers	27
8.3	Software Stack	27
9	Inference Request Flow	28
9.1	Serving Requirements	28

9.2 Capacity Sketch 28

10 Power 29

10.1 Per GPU Rack (32 GPUs / 4 servers) 29

10.2 BOOT Rack 29

10.3 STOR Rack 29

10.4 Facility 29

11 Cooling 30

12 Monitoring and Observability 31

12.1 GPU 31

12.2 Network 31

12.3 Application 31

13 High Availability 32

14 Expandability 33

15 Approximate Cost 34

16 Final Design Recommendation 35

17 Document Roadmap 36

18 Revision History 37

Status: Draft

Purpose: Production research AI platform for interactive inference of multi-trillion parameter MoE models

Scale: ~1,000 registered users, ~50 concurrent

Data retention: All inference retained **1 year** (research archive)

1 Design Goals

Priority	Goal
1	Interactive inference latency
2	Large model support (multi-trillion MoE)
3	Research workload flexibility
4	High availability
5	Expandability

This is **not** a maximum-throughput hyperscale training cluster. It is a manageable enterprise research inference platform.

2 Physical Summary

Item	Specification
GPUs	64× NVIDIA B200
GPU servers	8
GPUs per server	8
Racks (baseline)	4 (2× GPU + 1× bootstrap + 1× storage)
GPU rack density	32 GPUs/rack (4 servers)
Fabric	400GbE RoCEv2 rail-optimized (Arista 7060DX5)
NICs	8× 400GbE per server (1:1 with GPUs)
Orchestration	Kubernetes + NVIDIA GPU Operator
Primary workload	MoE LLM inference

3 Rack Layout and Facility Zones

3.1 How many racks?

Rack	Role	Required?
GPU-1	4× B200 servers (32 GPUs) + rail leaves share + OOB	Yes
GPU-2	4× B200 servers (32 GPUs) + rail leaves share + OOB	Yes
BOOT	DC bootstrap / management / control plane	Yes (minimum 1)
STOR	Hot model FS + 1-year inference object lake	Yes (1 is enough at this scale)
STOR-2	Archive growth / second failure domain	Optional (phase-2)
GPU-3	+32 GPUs expansion	Optional (phase-2)

Baseline footprint: 4 racks.

You do **not** need a second storage rack for day-1 750 TB–1.5 PB usable flash if nodes/appliances are dense. Add **STOR-2** when archive exceeds ~1.5–2 PB usable, you want rack-level failure isolation for the lake, or you separate “hot models” from “cold year archive” physically by policy.

3.2 Why a dedicated BOOT rack?

Keep bootstrap **off** the GPU power/cooling failure domain and free GPU RU for compute + rail optics.

Function	Lives in BOOT
K8s control plane (3 nodes)	Yes
PXE / image / iPXE / Metal3 or MAAS	Yes
DNS, DHCP, NTP, IPA/LDAP jump	Yes
NetBox (DCIM / IPAM / cabling SoT)	Yes
GitOps, registry, Vault/secrets	Yes
Monitoring / logging (Prometheus, Loki, ...)	Yes
Inference archive bus brokers (Kafka/Redpanda) if compact	Often here
CloudVision / NMS (optional)	Yes
Console servers, serial, crash carts path	Yes
Spines and/or border (optional co-locate)	Prefer here or row end
Storage data plane nodes	No → STOR
GPU workers	No → GPU-1/2

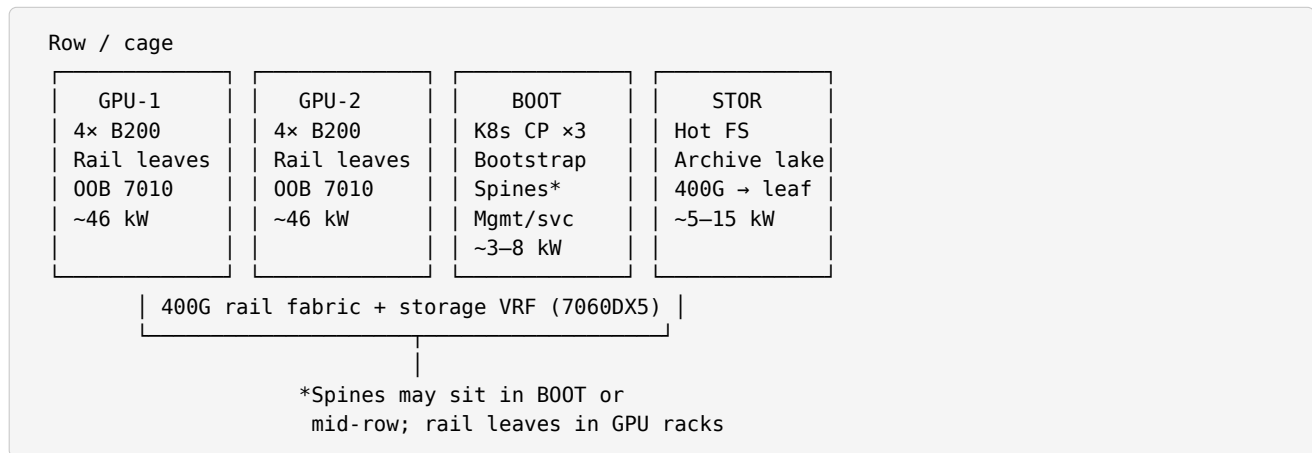
BOOT is **lightweight RU and kW** compared to GPU; one 42–48U rack is enough with headroom.

3.3 Why one STOR rack (usually)?

Tier	Day-1 size	Fits in one rack?
Hot models + landing	150–300 TB usable NVMe	Yes (3–6 nodes or 1–2 appliances)
Inference archive	≥750 TB usable (path 1.5–2 PB)	Yes with dense NVMe/object nodes or one scale-out appliance + shelves
Future 2 PB+ or dual-site style isola- tion	—	Plan STOR-2

Storage is **network-attached on 7060DX5**, not DAS to GPUs. Physical separation from GPU racks is good practice (blast radius, service windows); a *second* storage rack is capacity/isolation driven, not a hard topology requirement at 64-GPU scale.

3.4 Baseline elevation (logical)



* Prefer **spines in BOOT** (or dedicated network bay) so GPU racks stay power-dense for servers + busbars. Rail leaves stay close to GPU NICs (short DAC/AOC) → place in **GPU-1 / GPU-2**.

3.5 Per-rack contents (RU budget sketch)

Assumes ~42–48U racks; numbers are planning, not final elevation drawings.

3.5.1 GPU-1 / GPU-2 (each)

Contents	Qty	RU (approx)
B200 8-GPU servers (Dell XE9680)	4	24 (6U each)
Rail leaf 7060DX5-32 (share of 8)	3–5	3–5
OOB 7010TX-48	1	1
Patch / fiber manager	1–2	2–4
rPDU A/B	2	0–2
Headroom		reserve for optics density

Cable: 4 servers × 8×400G = **32× 400G** host links per GPU rack.

3.5.2 BOOT

Contents	Qty	RU (approx)
K8s control plane	3	3–6
Bootstrap / utility	2–4	2–8
Spines 7060DX5-64S	2	4
Optional service leaf ports / border	0–1	1–2
Redpanda/Kafka (if not on STOR)	3	3–6
Catalog DB / monitoring	2–4	2–8
OOB uplink aggregation / jump	1	1–2
Console / KVM	1	1
Headroom		large (growth, CVP, labs)

3.5.3 STOR

Contents	Qty	RU (approx)
Hot NVMe FS nodes or appliance	3–6 or 1–2	6–20
Archive object nodes / shelves	EC set	8–20
Storage 400G leaf attachment (or dual-home to fabric)	via GPU row leaves or local ToR ports	—
Storage BMC → OOB (extend from BOOT/GPU OOB)	—	—
Headroom		for +shelves before STOR-2

3.6 Decision summary

Question	Answer
Need BOOT rack?	Yes — at least one
Need extra racks for storage day-1?	One STOR rack is enough for hot + ≥ 750 TB archive
When add STOR-2?	$> \sim 1.5$ –2 PB usable, rack failure domain split, or policy isolation hot vs cold
Total baseline racks	4
Power domains	GPU-1/2 high-kW; BOOT + STOR normal enterprise power

4 Compute Nodes

4.1 GPU Server Platform

Standard platform: Dell PowerEdge XE9680 — 8 servers.

Item	Specification
Model	Dell PowerEdge XE9680
Chassis	6U rack server, air-cooled
GPU tray	NVIDIA HGX B200 (8× SXM)
Quantity	8 (4× GPU-1 + 4× GPU-2)

Alternates considered (not selected): NVIDIA MGX B200, HPE Cray XD, Supermicro HGX B200, Lenovo ThinkSystem AI. The rail/NIC/rack design is portable to any HGX B200-class 8-GPU node; BOM and RU drawings assume XE9680.

4.2 Per-Server Specification

Component	Specification
Chassis	Dell PowerEdge XE9680 (6U, air-cooled)
GPU	8× NVIDIA B200 (SXM, HGX B200)
GPU memory	~1.4 TB HBM3e
GPU compute	~1.4 PFLOPS FP8 (approx.)
CPU	2× Intel Xeon Scalable (4th/5th Gen)
CPU cores	96–128
System RAM	2 TB DDR5 ECC
Boot	2× M.2 1.92 TB NVMe RAID1 (BOSS-N1)
Local NVMe	8× 3.84 TB
Data network	8× 400GbE (1 NIC per GPU; rail-optimized)
NIC class	ConnectX-7 / BlueField-3 SuperNIC (PCIe Gen5 x16 slots)
GPU↔NIC	PCIe locality / GPUDirect RDMA; NIC paired to each B200
Management	1× 1GbE dedicated (iDRAC9)
Power	8–10 kW

4.3 Cluster Aggregate Capacity

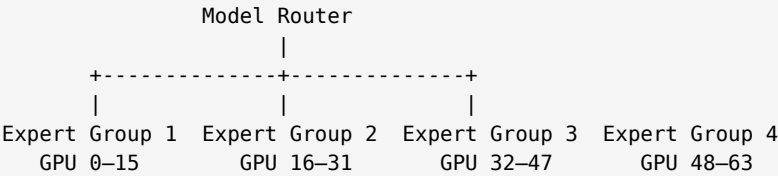
Resource	Amount
GPUs	64
HBM3e memory	~11.5 TB
CPU cores	~1,000
System RAM	16 TB

Resource	Amount
Local NVMe	~250 TB
GPU FP8 compute	~11–12 PFLOPS
Host 400G ports	64 (8 servers × 8 NICs)
Cluster host edge	25.6 Tbps

5 GPU Parallelism Model

For a ~3T MoE model, do **not** run one giant tensor-parallel job across all GPUs by default.

5.1 Logical Layout



5.2 Parallelism Dimensions

Dimension	Role
TP	Tensor parallel (within node / NVLink domain)
EP	Expert parallel (across nodes over fabric)
PP	Pipeline parallel (optional depth split)

5.3 Example Configurations

TP=8, EP=8, PP=1
TP=4, EP=16, PP=1

Exact values depend on model architecture, expert count, and latency targets.

6 Network Architecture

For MoE inference, the fabric is part of the computer.

Standard platform: Arista **7060DX5** series, EOS, optional CloudVision.

6.1 Host NIC Model — 1× 400GbE per GPU (required baseline)

Prior drafts assumed **2× 400G per server**. That is a common *shared-NIC* enterprise pattern and is **insufficient** as the baseline for this B200 MoE cluster.

Model	NICs / server	When used
Shared PCIe NIC	1-2× 400G	Light east-west; many general GPU clouds
Rail-optimized (this design)	8× 400G	1 NIC per GPU ; GPUDirect RDMA; MoE EP / NCCL

Baseline: each B200 has a **dedicated 400GbE** adapter (ConnectX-7 or BlueField-3 SuperNIC), PCIe-local to that GPU.

GPU0 — NIC0 — 400G — Rail-0 leaf

GPU1 — NIC1 — 400G — Rail-1 leaf

...

GPU7 — NIC7 — 400G — Rail-7 leaf

Why 1:1:

Reason	Effect
GPUDirect RDMA	GPU↔NIC without hairpin through another GPU's root complex
Avoid PCIe bottleneck	8 GPUs sharing 1-2 NICs saturates host I/O under EP
Rail scheduling	Same GPU index on every node lands on the same leaf (NCCL multi-rail)
Predictable EP	Expert-parallel all-to-all maps cleanly onto rails

Scope	Bandwidth
Per GPU	400 Gbps
Per server	3.2 Tbps (8×400G)
Cluster host edge	25.6 Tbps (64×400G)

6.2 Fabric Requirements

Feature	Requirement
Platform	Arista 7060DX5
NOS	Arista EOS
Link speed	400GbE (QSFP-DD)

Feature	Requirement
Host NICs	8× 400G per GPU server (1:1)
Topology	Rail-optimized leaf tier + spine
Transport	RoCEv2
Congestion	ECN, PFC, DCQCN / advanced congestion mgmt
Load balancing	ECMP + Dynamic Load Balancing (DLB); rail-aware where applicable
Overlay / control	BGP EVPN (VXLAN as needed)
Automation	eAPI / CloudVision
Tuning	AI / lossless Ethernet profile

6.3 Topology Rules

RoCEv2 lossless Ethernet does not require equal leaf and spine counts — but **rail count usually matches GPUs per node** (here 8 rails → 8 rail leaves).

What keeps MoE/EP well-behaved:

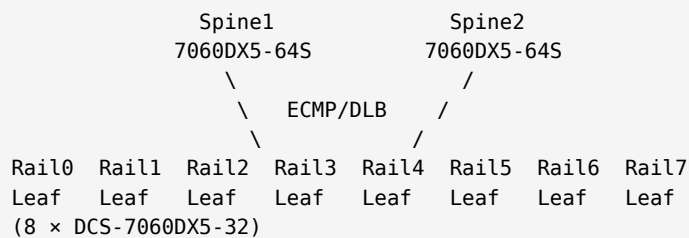
Requirement	Why it matters
1 NIC per GPU	Removes host NIC as the EP bottleneck
Rail-aligned wiring	GPU- <i>i</i> on all nodes → same rail leaf
Low / zero oversub leaf→spine	Cross-rail / non-rail traffic stays non-blocking
PFC + ECN + DCQCN	Lossless without pause storms
ECMP / DLB on spine tier	Spread cross-rail flows
Jumbo MTU end-to-end	Lower PPS
Buffer + queue telemetry	Microburst visibility

Spine count still follows bandwidth math (not “must equal leaf count” for lossless protocol reasons):

$$\Sigma \text{ leaf uplink BW} \geq \Sigma \text{ leaf downlink (host) BW} \quad \# \text{ for 1:1 non-blocking}$$

6.4 Topology — Rail-Optimized (default)

6.4.1 Default: 8 rail leaves + 2 spines



Role	Model	Qty	Notes
Rail leaf	DCS-7060DX5-32	8	One leaf per GPU index (rail 0–7)
Spine	DCS-7060DX5-64S (or 64E)	2	Cross-rail + storage/services

Host fan-in per rail leaf: 8 servers × 1 NIC = **8× 400G** downlinks (3.2 Tbps).

1:1 leaf→spine example: ≥ **8× 400G** uplink per rail leaf (e.g. 4×400G to each spine).

8 leaves × 8 uplinks = **64** leaf–spine 400G links → fits cleanly on **2× 64-port** spines with ports left for storage/border.

6.4.2 Optional: single-SKU square Clos

Use **8× DX5-32 spine** (or keep 2× 64S) if procurement wants all leafy SKUs. Still **8 rail leaves**; spine count is HA/BW, not “must be 8” for lossless.

6.4.3 Placement

Location	Gear
Rack 1	worker01–04 + share of rail leaves / OOB
Rack 2	worker05–08 + remaining rail leaves / OOB
Leaves may be split across racks; rail identity is logical , not “one rack owns four rails” only	

Cable so **GPU0 from every server** → **Rail0 leaf**, etc., regardless of rack.

6.5 Leaf Switches — Arista 7060DX5-32 (rail leaves)

Quantity: 8

SKU class: DCS-7060DX5-32

Feature	Value
Ports	32× 400G QSFP-DD
Breakout	up to 128× 100G
L2/L3 throughput	12.8 Tbps
Forwarding	~5.3 Bpps
Packet buffer	57 MB shared
Latency	from ~850 ns
Typical power	~289 W
Airflow	F-R or R-F (match rack)
HA hardware	1+1 PSU, N+1 fans
Routing	BGP EVPN, 128-way ECMP
AI features	DLB, advanced congestion, LANZ
QoS	PFC / ECN for RoCEv2

6.6 Spine Switches — 7060DX5-64 (qty 2)

Option	Model	Ports	Throughput	Buffer	RU	Notes
A1 (preferred)	DCS-7060DX5-64S	64× 400G QSFP-DD	25.6 Tbps	114 MB	2	Native 400G
A2	DCS-7060DX5-64E	32× 800G → 64× 400G	25.6 Tbps	114 MB	1	800G path

Feature	Value
ECMP	up to 128-way
Latency	from ~850 ns
Typical power	~489 W (64S) / ~548 W (64E)
Airflow	Front-to-rear
HA hardware	1+1 PSU, N+1 fans

Default procurement: 8× DX5-32 rail leaf + 2× DX5-64S spine.

6.7 Server Connectivity (per GPU server)

Dell PowerEdge XE9680 (HGX B200, 8 GPUs)

- └ GPU0 — NIC0 — 400G — Rail-0 leaf
- └ GPU1 — NIC1 — 400G — Rail-1 leaf
- └ GPU2 — NIC2 — 400G — Rail-2 leaf
- └ GPU3 — NIC3 — 400G — Rail-3 leaf
- └ GPU4 — NIC4 — 400G — Rail-4 leaf
- └ GPU5 — NIC5 — 400G — Rail-5 leaf
- └ GPU6 — NIC6 — 400G — Rail-6 leaf
- └ GPU7 — NIC7 — 400G — Rail-7 leaf

Within the node, **TP** prefers **NVLink**; NICs carry **inter-node EP/PP**, storage, and control that is mapped onto data NICs. Do not design as if 8 GPUs share two uplinks.

Scope	Bandwidth
Per GPU / NIC	400 Gbps
Per server	3.2 Tbps
Cluster host edge	25.6 Tbps (64× 400G)

6.8 Port Plan (Planner Baseline)

Per rail leaf (7060DX5-32):

Use	Ports	Speed
GPU rail downlinks (8 workers × GPU- <i>i</i>)	8	400G
Spine uplinks (1:1 → ~8 total)	8	400G
Storage / services (subset of leaves)	0–4	400G
Spare / growth	remainder	400G

Leaf front-panel: 8 down + 8 up = 16 ports used for pure compute fabric; 32-port SKU has headroom.

Per spine (7060DX5-64S):

Use	Ports (example)	Speed
Rail leaf uplinks (8 leaves × 4)	32	400G
(if 8 uplinks/leaf: 8×8)	64	400G
Storage cluster	2–8	400G
Border / DCI / services	2–4	400G
Spare	remainder	400G

With **8 uplinks per leaf**, two 64-port spines are fully subscribed on leaf-facing ports if split 32+32 — size storage on extra spines/ports or reduce leaf uplinks only if accepting oversub.

6.9 Optics and Cabling

Link	Preferred media
Server ↔ leaf (same rack)	400G-DR4 / DAC or AOC as length allows
Leaf ↔ spine (in-row / adjacent)	400G-DR4 or FR4 per plant standards
Connector	QSFP-DD on 7060DX5-32 / 64S
Spares	20% optics/cables minimum

Exact optics SKUs finalized in docs/bom.md and docs/network.md.

6.10 RoCEv2 / AI Ethernet Profile (Arista EOS)

Required fabric behaviors for MoE all-to-all / EP traffic:

Control	Setting direction
Priority flow control	Lossless queue for RoCE PFC priority
ECN	Enabled on RoCE queue; mark before drop
DCQCN	Host CNP reaction; switch ECN marking thresholds tuned
DLB	Prefer Dynamic Load Balancing over static hash for large flows
MTU	Jumbo (e.g. 9000) end-to-end
ECMP	Max-path ≥ leaf uplink count; resilient hashing
Telemetry	LANZ / queue depth; drop & ECN counters scraped

6.11 Out-of-Band / Management Network

Separate from the 400G RoCE fabric. **Never** carry GPU data-plane traffic.

Standard platform: Arista **7010TX-48** (7010X series), EOS.

6.11.1 Platform

Item	Spec
Model	DCS-7010TX-48 (AC) or DCS-7010TX-48-DC
Form factor	1RU

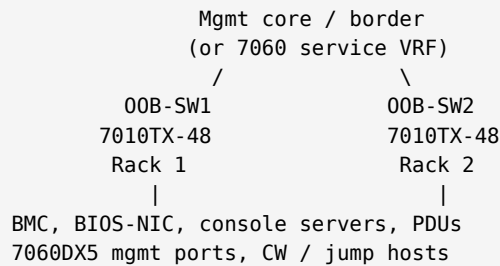
Item	Spec
Access ports	48× 10/100/1000BASE-T (RJ45)
Uplinks	4× SFP28 (1/10/25GbE)
Throughput	296 Gbps / ~220 Mpps
Buffer	4 MB shared
Power	1+1 redundant PSU
Fans	1+1 hot-swap; reversible airflow
NOS	Arista EOS (same family as 7060DX5)

Related SKUs: 7010TX-48C / 7010TX-48C-DC if copper/console variant needed per quote.

6.11.2 Quantity and placement

Role	Qty	Placement
OOB leaf	2	1 per GPU rack (baseline)
Optional 3rd	1	Management rack (K8s + storage BMCs) if port pressure

Two OOB switches are dual-homed uplink or MLAG pair so a single OOB switch failure does not black-hole BMC access.



6.11.3 What attaches to 7010TX-48

Endpoint	Speed	Notes
GPU server BMC (iDRAC)	1G	Always-on lights-out
GPU server mgmt NIC (mgmt0)	1G	OS install / PXE (VLAN 10)
K8s control-plane mgmt	1G	Dedicate ports; not RoCE NICs
7060DX5 Management1	1G	Out-of-band EOS mgmt + ZTP
Storage / PDU / serial consoles	1G	As present
CloudVision / jump / IPMI tools	1G	Ops plane

6.11.4 Port budget (per rack, 4 GPU servers)

Use	Ports (approx.)
4× GPU BMC	4
4× OS mgmt NIC (optional)	0–4
2× leaf 7060DX5 mgmt	2

Use	Ports (approx.)
1× spine share or local	0–1
PDU / rPDU	2–4
Console server	1
Spares	remainder of 48

48 ports per rack is comfortable at this scale.

6.11.5 Uplinks

Item	Spec
Ports	1–2× 10G (or 25G) SFP28 per 7010TX-48
Target	Redundant path to mgmt/core or services block
HA	MLAG pair between the two 7010TX-48 (peer-link 2× SFP28) — design in docs/network.md §4

OOB is **loss-tolerant best-effort** (no RoCE PFC). Do not enable lossless QoS on 7010TX-48.

6.11.6 Security

Control	Requirement
Isolation	Dedicated VRF / VLAN; no route to RoCE data VRF except controlled jump
Access	MFA jump host; no public BMC
ACLs	Allow only mgmt subnets → BMC/IPMI/SSH/HTTPS
Logging	EOS AAA + syslog/telemetry to ops stack

7 Storage

Inference is **not** a training I/O path. Storage must be fast enough to **load multi-TB MoE checkpoints**, hold versions, and support research artifacts — without competing with RoCE expert traffic incorrectly, and without needing an all-flash training forge.

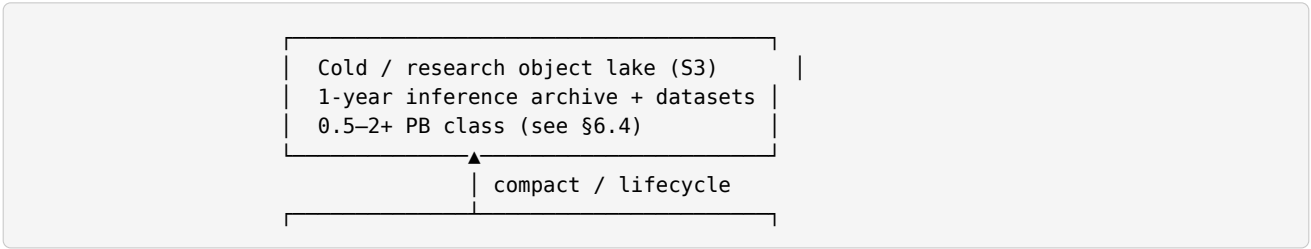
7.1 Role of Storage in This Cluster

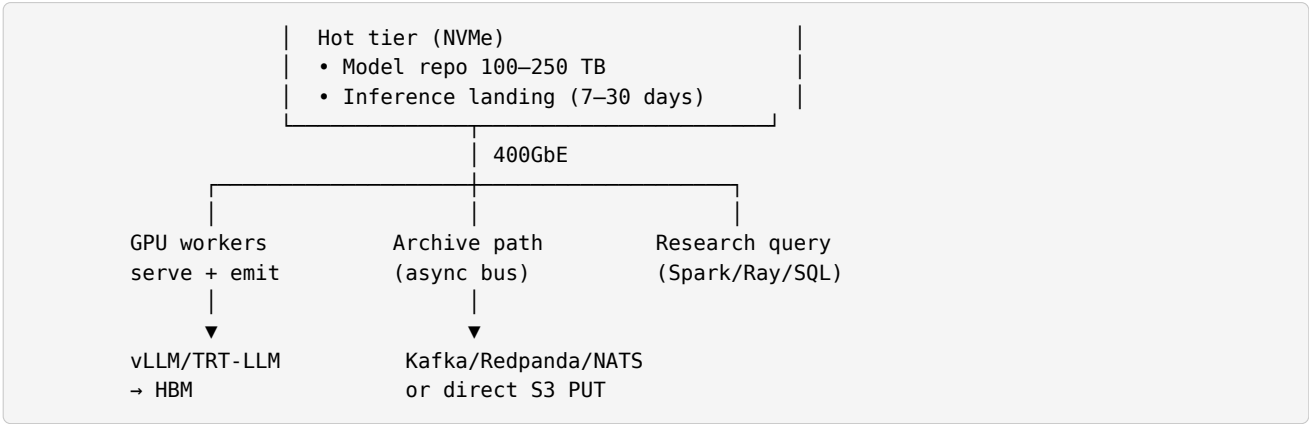
Path	When it runs	Bandwidth need	Latency sensitivity
Model cold start / roll-out	Deploy, restart, scale, canary	High burst (tens of GB/s cluster)	Startup time only
Steady inference	After weights in HBM/DRAM	Near-zero for weights	Tokens from GPU HBM
Inference archive (1 year)	Every request (async)	Sustained write + research scan	Write path must not block TTFT
KV / prefix cache spill (optional)	Long context / multi-tenant	Medium, local-first	Moderate
Datasets / eval / embeddings	Research jobs (Ray)	Medium sequential	Low-medium
Ops metrics / traces	Always	Low	Low

Design principles:

- Weights** live on hot shared NVMe; steady-state path is GPU HBM.
- All inference** (prompts, completions, metadata) is retained **≥ 1 year** for research.
- Archive write is **async / sidecar** — never on the critical token path.

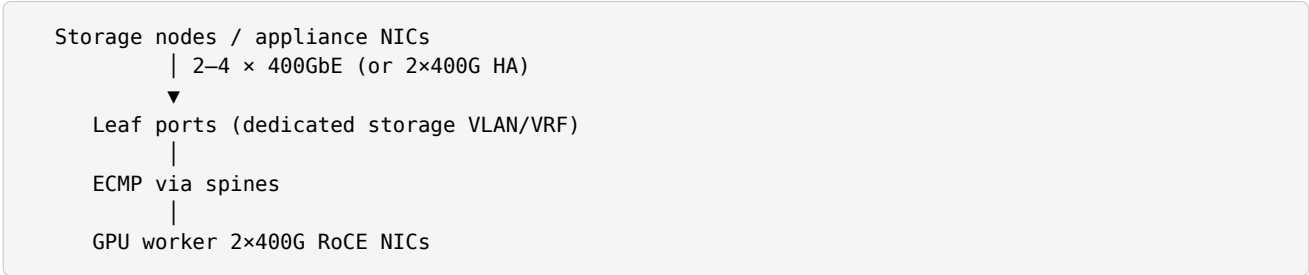
7.2 Logical Architecture





7.3 Integration with the Network

Storage attaches to the **same Arista 7060DX5 data fabric** as GPU servers (not the 7010TX-48 OOB).



Item	Spec
Fabric	7060DX5 leaf-spine
Host mount path	GPU NICs (data plane), not BMC/OOB
Isolation	Dedicated storage VRF or VLAN; QoS separate from pure RoCE EP class if needed
Transport	NFS/RDMA, Weka/BeeGFS/Lustre client, or S3 over TCP/HTTP — product-dependent
Oversubscription	Storage uplinks sized so model load does not starve EP if sharing queues; prefer separate DSCP/PFC priority for storage vs RoCE bulk if mixed

BMC / storage controllers → 7010TX-48 OOB only.

7.4 Capacity and Performance Targets

7.4.1 Hot tier (models + short landing)

Metric	Target
Usable NVMe	100–250 TB models + 20–50 TB inference landing
Single large MoE checkpoint	5–40 TB compressed/sharded per version
Concurrent model versions	3–5 online
Aggregate model read	≥ 40 GB/s (stretch 80+)
Single-node model read	≥ 10–20 GB/s

7.4.2 Inference archive tier (1-year retention) — required

Policy: retain **every** inference exchange for **365 days** (research cluster default). Extendable later; deletes only via explicit policy after retention.

Recorded per request (minimum):

Field	Required
request_id, trace_id, timestamp (UTC)	Yes
Tenant / user id (pseudonymized if needed)	Yes
Model id + version + serving revision	Yes
Full prompt / messages (as served)	Yes
Full completion tokens (as returned)	Yes
Sampling params (temp, top_p, max_tokens, seed)	Yes
Token counts in/out, TTFT, TPOT, status, error	Yes
Router / expert stats (if available)	Research-preferred
Logprobs / top-k (optional, large)	Opt-in per project
Tool calls / RAG citations	When used
Content hash + schema version	Yes

Do not store on the critical path: large binary attachments stay as object pointers; text payload is first-class.

7.4.3 Capacity planning model

Planning assumptions from cluster goals (~50 concurrent, ~32k context class, research use):

Sce-nario	What is stored	Est. avg sustained	1-year retained (after compression)
Lean	Text in/out + metadata; multi-turn delta or single final turns; zstd/parquet	~0.5–2 TB/day raw → compress ~3–5×	~50–150 TB
Stan-dard re-search (de-fault)	Full prompts+comple-tions each call; light metadata; parquet+zstd; some history du-plication	~2–8 TB/day raw → compress ~3–5×	~200–600 TB
Full fi-delity	1. logprobs / n-best / rich traces; frequent long 32k contexts	~10–40 TB/day raw	~1–3 PB

Baseline procure for this program: size the **object/archive lake for ≥ 750 TB usable** (covers standard research + headroom), with a growth path to **1.5–2 PB** without re-architecture. Revisit after 30–90 days of measured bytes/request.

Quick sanity math (standard):

~2 req/s average × 150 KB JSON-equivalent/request
 ≈ 25 GB/day uncompressed text-ish payload
 × research overhead / multi-turn / indexes ≈ 3–10×
 → hundreds of TB/year compressed columnar is realistic

Peak token generation (~1k tok/s) is **not** the archive size driver; **full prompt context logged per turn** is.

7.4.4 Archive performance targets

Metric	Target
Ingest durability	Multi-AZ or erasure-coded object; no single disk loss of year
Write path	Async; p99 archive lag ≤ 60s under normal load
Impact on TTFT	None (buffer + drop-oldest on extreme backlog with alert — research may forbid drop; then backpressure only non-interactive jobs)
Query	Partition by day/model/tenant; scan last 7d interactive; full-year batch OK
Immutability	WORM-optional; default append-only prefixes + lifecycle

7.5 Node-Local Storage (already on GPU servers)

Each GPU server has **8× 3.84 TB NVMe** (~30 TB raw). Use as:

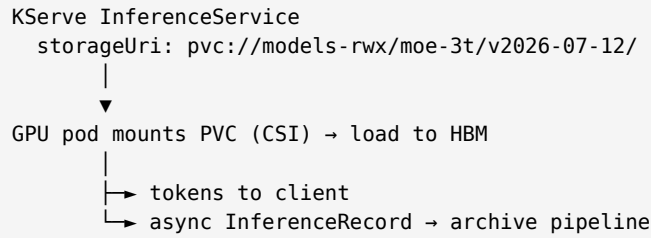
Mount	Purpose
/var/lib/kubelet / containerd	Images, ephemeral
/scratch or hostPath	Shard download cache, compile cache
Optional KV offload dir	Framework spill (if enabled)

Not primary model catalog. Rebuild-from-shared is the DR story for local cache.

7.6 Kubernetes Integration

Mechanism	Use
CSI driver (NFS, Weka, CephFS, Dell/Pure, etc.)	RWX for model weights + shared datasets
PVC models-rwx	Mounted read-mostly on all GPU pods
initContainer / Job	Stage weights → local NVMe if faster second start
KServe StorageUri	pvc:// or s3:// → runtime load
Inference logger	Sidecar or gateway plugin → bus/object (see §6.7)
Image is not the model	Do not bake multi-TB weights into images

Example **serving** path:



7.7 Inference Capture and 1-Year Archive Pipeline

Research requirement: **100% of inference traffic retained 1 year.**



Stage	Tech direction	Notes
Emit	OpenTelemetry gen-AI attrs or custom JSON schema	Stable <code>schema_version</code>
Transport	Kafka/Redpanda/NATS or direct multipart S3	Prefer bus if multi-consumer
Format	Parquet (zstd) primary; JSONL only for debug	Columnar = year-scale scans
Partition	<code>day / model_version / tenant</code>	Prune queries
Index	Lightweight catalog DB (<code>request_id</code> → object key)	Point lookup without full scan
Retention	365 days default lifecycle rule	Legal hold flag available
Access	Read via research VPC/IAM; GPU path write-only key	Segregate credentials

Gateway vs sidecar: prefer **API gateway or dedicated logger service** once, so multipod TP/EP shards do not multiply records. If logging at engine, **dedupe on request_id**.

PII / policy: research cluster still needs classification hooks (blocklist, user opt-out, secret scrub) before durable write; rejected payloads stored as tombstone + reason code when policy denies full text.

7.8 Data Layout

Models (hot FS):

```
/models/
  moe-3t/
    v2026-07-01/
    v2026-07-12/
      config.json
      shards...
      CHECKSUMS
    embeddings/
    adapters/
  /datasets/
  /eval/
  /artifacts/
```

Inference archive (object):

```
s3://inference-archive/
  year=2026/month=07/day=18/
    model=moe-3t/ver=v2026-07-12/
      part-000.parquet
      part-001.parquet
    _catalog/
      request_id embeddings or SQLite/Postgres export
    _schema/
      vl.json
```

Promote models with immutable dirs + atomic current. Archive is append-only by day.

7.9 Platform Options

Split **hot model FS** from **year-long inference lake** (can be one product with two pools, or two products).

Tier	Preferred options	Role
Hot models	Weka / BeeGFS / Lustre / FlashBlade-class NFS	Checkpoint load RWX
Inference archive	S3-compatible: Ceph RGW, MinIO, Vast, Pure, cloud-adjacent	1-year lake, parquet
Bus (optional)	Redpanda / Kafka	Fan-out to archive + online eval
Catalog	Postgres / OpenSearch	request_id lookup, audit

Default recommendation:

Need	Choice
Model load	Parallel FS or appliance NFS on NVMe (100–250 TB)
1-year inference	S3 API object lake $\geq$ 750 TB usable , erasure coding, lifecycle 365d
Research query	External table / Ray datasets over parquet prefixes

7.10 Reference Hardware Footprint

Component	Qty / size (indicative)
Hot NVMe model (+ landing)	3–6 nodes/appliance, 150–300 TB usable
Archive object nodes or appliance	EC set, $\geq$ 750 TB usable (path to 2 PB)
400GbE to 7060DX5 leaves	2–4 ports per storage node/pair
Archive ingest brokers (if used)	3 VMs/nodes on mgmt or storage rack
OOB	All BMCs $\rightarrow$ 7010TX-48

Exact BOM: docs/storage.md / docs/bom.md.

7.11 Failure and Lifecycle

Event	Behavior
Hot FS node loss	Models still load (repl/EC); slower start
Archive node loss	No loss of committed objects (EC/repl)
Bus backlog	Alert; expand consumers; do not block token stream unless policy=strict
Fabric leaf loss	ECMP dual-home storage + GPU
Bad model promote	Immutable versions; roll back InferenceService
GPU wipe	Re-pull models; local cache disposable
Day-366	Lifecycle delete or legal-hold keep
Steady inference vs hot FS outage	Live QPS OK if weights loaded; new starts fail
Archive write outage	Live QPS OK; gap alarms; replay from buffer if any

7.12 What Storage Does *Not* Do Here

- Not a substitute for HBM
- Not the MoE expert all-to-all path (GPU RoCE/NVLink)
- Not on 7010TX-48
- Not training-scale continuous multi-TB/s checkpoint writes
- Inference archive is **not** mounted as latency-critical RWX on GPU pods

7.13 Summary Integration Map

Plane	Switch / system	Storage role
Data fabric	7060DX5	Models + archive ingest + research scans
OOB	7010TX-48	BMCs only
Hot FS	CSI RWX	Checkpoints

Plane	Switch / system	Storage role
Object lake	S3 API	1-year inference + artifacts
Bus	Kafka/Redpanda (opt)	Reliable async capture
Local NVMe	GPU server	Cache / emit buffer
GPU HBM	B200	Steady-state weights

8 Kubernetes Architecture

8.1 Management Nodes

Quantity: 3 (HA control plane)

Component	Spec
CPU	32 cores
RAM	256 GB
Storage	NVMe

8.2 GPU Workers

worker01 ... worker08
each: 8× B200

8.3 Software Stack

Kubernetes
└─ NVIDIA GPU Operator
 └─ CUDA
 └─ TensorRT-LLM / vLLM
 └─ KServe
 └─ Ray (research / batch)

Layer	Role
Kubernetes	Scheduling, HA, multi-tenancy
GPU Operator	Driver, device plugin, DCGM, MIG (if used)
vLLM / TensorRT-LLM	High-performance serving
KServe	Model serving CRDs, canary, autoscaling hooks
Ray	Research jobs, eval, offline batch

9 Inference Request Flow

User

→ API Gateway

→ Authentication / Authorization

→ Inference Scheduler

→ Model Server

→ GPU Cluster

→ Response Tokens

9.1 Serving Requirements

Attribute	Target direction
Latency	Interactive (TTFT + TPOT tuned)
Context	Up to 32k (initial planning assumption)
Concurrency	~50 simultaneous users
Throughput	~1,000 tokens/sec aggregate

9.2 Capacity Sketch

50 concurrent users × 20 tokens/sec/user ≈ 1,000 tokens/sec

A 64× B200 cluster is in the correct range for this profile on a large MoE, subject to model/kernel tuning.

10 Power

10.1 Per GPU Rack (32 GPUs / 4 servers)

Component	Watts
4 GPU servers	~40 kW
Rail leaves share + optics	~2–3 kW
OOB / TOR share	~0.5 kW
GPU rack total	~43–46 kW

10.2 BOOT Rack

Component	Watts
K8s + bootstrap + utilities	~2–4 kW
Spines (2× DX5-64S)	~1–1.5 kW
Bus / monitoring	~1–2 kW
BOOT total	~4–8 kW

10.3 STOR Rack

Component	Watts
Hot FS + archive NVMe (day-1)	~5–12 kW
Growth to ~1.5–2 PB	up to ~15 kW class
STOR total (day-1)	~5–15 kW

10.4 Facility

Item	Spec
Cluster IT load (baseline 4 racks)	~100–115 kW
Available facility power	130–150 kW recommended (headroom + STOR growth)
GPU rows	High-density 415 VAC, A/B; ~50 kW/rack class
BOOT + STOR	Standard density OK; A/B feeds
Feeds	A/B redundant all racks

11 Cooling

Tier	Approach
Minimum	Hot-aisle containment + high-airflow CRAC
Preferred	Rear-door heat exchangers
Future	Direct liquid cooling (DLC)

Plan rack and facility layout so DLC retrofit is possible.

12 Monitoring and Observability

12.1 GPU

- NVIDIA DCGM
- GPU utilization
- HBM usage
- ECC errors
- NVLink health
- Power / thermal

12.2 Network

- Interface errors
- ECN marks
- Packet drops
- RoCE congestion counters
- Fabric latency / loss

12.3 Application

- Tokens/sec
 - Time-to-first-token (TTFT)
 - Time-per-output-token (TPOT)
 - Queue depth
 - GPU occupancy / KV-cache utilization
 - Error rate / timeouts
-

13 High Availability

Domain	Approach
Control plane	3-node Kubernetes HA
Fabric	8×400G/node rail leaves (7060DX5-32 ×8), dual spine (7060DX5-64) ECMP + DLB
Power	A/B feeds
Model serving	Multi-replica where parallelism allows; rolling updates via KServe
Storage	Redundant NVMe pool / erasure or replication per product

Failure domains should align with racks and leaf pairs so a single leaf or rack loss does not take the full model offline if EP groups are placed carefully.

14 Expandability

Path	Notes
+1 GPU rack (32 GPUs)	Add rail leaf capacity as needed, extend EVPN, join workers
+server within GPU rack	Power/cooling/RU headroom first
Storage growth in STOR	Add NVMe nodes/shelves until RU/power full
+ STOR-2	When usable archive >~1.5-2 PB or rack-level failure split
Multi-model	Partition expert groups / namespaces by team

Initial design: **4 racks** with clear expansion to 5-6 (GPU-3, STOR-2) without redesigning fabric roles.

15 Approximate Cost

Category	Estimate
64× B200 GPUs	\$4–6M
GPU servers	\$1–1.5M
Network fabric	\$1–1.5M
Storage	\$500k
Rack / power / cooling	\$500k
Total	\$7–10M

Costs are order-of-magnitude planning numbers only; obtain vendor quotes for procurement.

16 Final Design Recommendation

For a research organization running interactive multi-trillion MoE inference:

Layer	Choice
Compute	64× B200 / 8× Dell PowerEdge XE9680 / 2 GPU racks
Facility	4 racks: GPU-1, GPU-2, BOOT, STOR
Network	8×400G/node rail fabric: 8× DX5-32 + 2× DX5-64S; 7010TX-48 OOB
Storage	100TB+ NVMe model repository
Platform	Kubernetes + GPU Operator + vLLM/TensorRT-LLM + KServe + Ray
Ops	DCGM + EOS/LANZ fabric telemetry + token-level SLOs

This is the scale at which the system behaves like a real AI supercomputer while remaining operable by an enterprise infrastructure team.

17 Document Roadmap

Follow-on specs (to be written next):

1. `docs/bom.md` — bill of materials and SKUs
 2. ~~`docs/network.md`~~ — **done** (ZTP day-0, numbered eBGP underlay, RoCEv2 lossless profile)
 3. ~~`docs/bootstrap.md`~~ — **done** (BOOT seed, Metal3/CAPI, Flux platform) + `bootstrap/`
 4. ~~`docs/netbox.md`~~ / ~~`docs/cabling.md`~~ — **done** (NetBox DCIM/IPAM SoT + cabling export)
 5. `docs/k8s.md` — day-2 cluster ops, tenancy, upgrades
 6. `docs/storage.md` — model repo layout and performance targets
 7. `docs/facility.md` — rack elevations, power, cooling
 8. `docs/slo.md` — latency/throughput SLOs and capacity model
 9. ~~`docs/build-guide.md`~~ — **done** (physical build: rack, power, cable, bring-up, burn-in)
 10. ~~`docs/overlay.md`~~ — **done** (EVPN/VXLAN overlay: vrf **storage** + vrf **edge**, border, API ingress)
 11. ~~`docs/serving.md`~~ — **done** (Kimi K2 Thinking deploy, inference scheduler, API gateway/auth, capture wiring)
-

18 Revision History

Version	Date	Notes
0.1	2026-07-18	Initial specification
0.2	2026-07-18	Network standardized on Arista 7060DX5 (DX5-32 leaf, DX5-64S spine)
0.3	2026-07-18	Clarified lossless $\neq$ equal leaf/spine; added square Clos Option B
0.4	2026-07-18	OOB standardized on Arista 7010TX-48 (2×, 1 per rack)
0.5	2026-07-18	Storage integration: fabric, CSI/KServe, tiers, targets
0.6	2026-07-18	1-year inference archive lake (≥ 750 TB usable, async capture)
0.7	2026-07-18	Host NIC 8×400G (1:1 GPU); rail-optimized 8 leaf + 2 spine
0.8	2026-07-18	Facility: 4 racks (2× GPU + BOOT + STOR); STOR-2 optional
0.9	2026-07-18	Bootstrap stack: seed + Metal3/CAPI + Flux platform (bootstrap/, docs/bootstrap.md)
1.0	2026-07-18	NetBox SoT: inventory, IPAM, cabling guide (bootstrap/netbox/, docs/netbox.md, docs/cabling.md)
1.1	2026-07-19	GPU servers standardized on Dell PowerEdge XE9680 (8×, 6U)
1.2	2026-07-19	Network: ZTP day-0 (OOB bench ZTP $\rightarrow$ relay ZTP), eBGP underlay, RoCEv2 lossless profile (docs/network.md); OOB = MLAG pair; +mgmt0/Ma1/peer cabling (169 cables)
1.3	2026-07-19	EVPN/VXLAN overlay (vrf storage/edge, border — docs/overlay.md); Kimi K2 Thinking deployment + inference scheduler + API access (docs/serving.md, bootstrap/platform/apps/{models,api})